

Hussain Mustansir

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Computer Science graduate with experience building backend systems, REST APIs, and cloud infrastructure across two software engineering internships. Looking to grow as a backend or full-stack engineer on products that ship fast and scale.

Education

Bachelor of Science in Computer Science

Habib University, Karachi, Pakistan | 2022 – 2026

Technical Skills

- Frontend & Web: React.js, Next.js, JavaScript/TypeScript, HTML/CSS, Responsive UI/UX Design Implementation
 - Programming & Backend: JavaScript (Node.js, Express.js), Python, Django, REST/FAST API Integration
 - Cloud & DevOps: Docker containerization, CI/CD pipelines (GitHub Actions), AWS/Azure concepts
 - Databases: MongoDB, PostgreSQL, MS SQL, schema design, API data integration
 - Tools & Systems: Git/GitHub, Postman, JMeter, Selenium, ClickUp, JIRA
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Experience

- **Software Engineer | Observeri Technologies | Dec 2025 - Present**
 - Built and maintained backend services for GRC Platform, using Node.js and Express.js.
 - Collaborated and contributed to frontend development and handled UI integrations using React.js.
 - Built a centralized API audit logging system to track every CRUD operation across a multi-tenant backend.
 - Developed a vulnerability scanner using ZAP with persisting results to the multi-tenant vulnerability register.
- **Software Engineer Intern - Platform | Data Science Dojo | March 2026 – June 2026**
 - Built and maintained backend services for Ejento AI, using Python, Django.
 - Deployed and managed AI model configurations through Azure AI Foundry and API Management (APIM).
 - Designed REST APIs following OpenAPI standards, including versioning, error handling, and documentation.
 - Optimized API and database performance through query optimization, caching, and distributed system design.

Academic Projects & Research Publication

- **Karavaan | Spursol | Final Year Project**
 - Built a Node.js/Express REST API for different components with a PostgreSQL database (30+ tables).
 - Integrated messaging feature with [Socket.io](#), google mapping services and Expo notifications for in-app/push alerts.
 - Built auth flows: registration, login, Google OAuth, OTP (AWS SES) with secure token handling.
 - Implemented event discovery and recommendations using Recombee AI engine.
- **[Prediction Model For Legal Judgments](#) | Research Publication**
 - Built an AI model using Transformer-based NLP models (BART, T5 Base, Distil GPT-2, Facebook-opt-125m) to predict criminal case judgments under Pakistani law and evaluated accuracy using ROUGE-1 scores.
 - Processed and trained on judgments using Python using libraries BeautifulSoup, Tesseract, and Pandas.
- **Multilingual Resume Retrieval, Job-Skill Matching + Classification | Research Publication**
 - Built four retrieval pipelines to match resumes to job descriptions using bi-encoders, BM25, and cross-encoder.
 - Fine-tuned BGE-Large with hard-negative mining for resume-to-job matching, and tested a TripletLoss pipeline.
 - Built a job-skill matching system by fine-tuning JobBERT embeddings and combining them with ESCO knowledge graph signals, then used a cross-encoder to classify each skill as core or contextual.
- **IoT Smart Home System with Serverless Data Pipeline | Course Project**
 - Built end-to-end IoT data pipeline: ESP32 sensor nodes (temperature, humidity, light, RFID) → MQTT over TLS → AWS IoT Core → S3 data lake → Amazon Athena (SQL engine) → Power BI dashboards.
 - Implemented cloud command loop using AWS Lambda for RFID-based authentication and solenoid lock actuation